

Parth Deshpande

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Education

VIT Bhopal University, Bhopal

June 2023 – June 2027

Bachelor of Technology in Computer Science & Engineering (AI&ML)

CGPA: 9.0/10

Govt. Excellence H. S. School, Betul

2023

Class 12th Board: 94%

Class 10th Board: 100%

Technical Skills

Languages: Python, C++, JavaScript (ES6+), SQL, Java

AI & Machine Learning: PyTorch, TensorFlow, Hugging Face, NLP, OpenCV, LLMs

Web Development: React.js, Node.js, Express, HTML5, CSS3, RESTful APIs

Databases: PostgreSQL, MongoDB, MySQL, FAISS (Vector DB), Firebase

Developer Tools & Cloud: Git, Docker, AWS (EC2, S3), MLflow, Postman, VS Code, Linux

Projects

Vector-Database: High-Dimensional Vector Search Engine

Jun 2026 – Jul 2026

Technologies: C++, HNSW, KD-Tree, cpp-httplib (REST API), Ollama, RAG

- Engineered a vector database from scratch in C++, implementing HNSW, KD-Tree, and Brute-Force search side-by-side across 3 configurable distance metrics (cosine, Euclidean, Manhattan), benchmarking $O(\log N)$ approximate retrieval against exact $O(N \cdot d)$ baselines.
- Built a full RAG pipeline integrating local LLM inference (Ollama `nomie-embed-text` + `llama3.2`) with the custom HNSW index, chunking documents into overlapping 250-word segments and streaming token-level answers over a REST API.
- Designed a real-time PCA-based 2D visualization dashboard to project 16D/768D embeddings and expose semantic clustering across demo vectors and user-uploaded documents.

Semantic Code Similarity: NLP-Based Plagiarism Detector

Feb 2026 – Mar 2026

Technologies: Python, Word2Vec, GloVe, NLP, scikit-learn, Streamlit

- Built an NLP-based plagiarism detector that compares Python source code semantically rather than syntactically, using tokenization, identifier normalization, and mean-pooled Word2Vec/GloVe embeddings.
- Implemented dual embedding backends and configurable cosine-similarity thresholds to flag code obfuscated via variable renaming, comment edits, or reformatting; deployed an interactive Streamlit UI for two-snippet and multi-file batch analysis.
- Designed a pairwise similarity heatmap for folder-level plagiarism scans across an entire codebase in a single pass.

Subway Surfers Virtual Play: Real-time HCI System

Dec 2025 – Jan 2026

Technologies: Python, OpenCV, MediaPipe, PyAutoGUI

- Engineered a real-time Human-Computer Interaction system controlling Subway Surfers through live body pose detection, eliminating keyboard input entirely via gesture-to-keypress mapping.
- Designed a calibration and landmark-tracking pipeline using MediaPipe Pose to classify 5 game actions from shoulder and hip landmark geometry.
- Structured the project into modular utility files with dedicated test scripts per gesture, enabling isolated debugging and rapid iteration.

Achievements

- Selected for Amazon ML Summer School 2026 — top 2.2% (~3,000) of 1.34 lakh+ registrations
- Attained Pupil Rank (Max Rating: 1409) on Codeforces
- Mastered DSA — 900+ problems on LeetCode, Codeforces & CodeChef
- Finalist, HackTU 6.0 — top team among 3,500+ students, 800+ teams

Certifications

- Oracle Agentic AI Certified Foundations Associate
- 5-Day AI Agents: Intensive Vibe Coding Course With Google — Kaggle
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Applied Machine Learning in Python — University of Michigan (Coursera)
- Cloud Computing (NPTEL) — IIT Kharagpur